

Nasim Mahmud Nayan

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SUMMARY

AI Solution Architect with 4+ years of experience designing and delivering AI systems across healthcare, education, and development. Advancing responsible AI use and AI governance across programmes at BRAC, one of the world's largest development organisations, with a focus on supporting frontline work and turning programme data into evidence that decision-makers can act on. Published researcher in fair and explainable AI with 12 papers and 174 citations, now working on trust and interpretability in large language models. Peer reviewer for two international journals.

SKILLS

AI Strategy & Governance	AI strategy, use-case prioritisation, responsible AI, AI governance, fairness & explainability, vendor assessment
AI & GenAI	LLMs, RAG, AI agents, NLP, Computer Vision, Predictive Modelling, Recommender Systems
Architecture & Tools	Enterprise AI architecture, data pipelines, model lifecycle, PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, Cloudera, MLflow, Docker, FastAPI, vector databases, PostgreSQL/pgvector
Leadership	Stakeholder management, cross-functional delivery, vendor coordination, leadership reporting, technical mentoring
Languages	Python, C++, C

WORK EXPERIENCE

BRAC | *Deputy Manager, AI & Machine Learning* | *Central Data Team* May 2026 – Present

- Owning the AI and machine learning roadmap and exploring AI opportunities across BRAC's programmes and enterprises, so that frontline colleagues get support in their daily work where the need is real.
- Turning BRAC's existing data into a base for quick, evidence-led decisions, using AI and NLP to connect it to BRAC's five-year strategic goals as part of a targeting dashboard for leadership.
- Contributing to AI governance across BRAC by shaping policy on responsible AI use, data protection for AI products, and the standards a system must meet before it reaches frontline work at population scale.
- Overseeing AI products built by programme teams and specialist vendors across the model lifecycle, including a legal advisory chatbot for SELP and an agricultural advisory tool with the Social Innovation Lab, reviewing the architecture and problem fit and advising where the approach needs to change.

Solynta Limited | *AI Engineer* Dec 2025 – Apr 2026

- Built a RAG-based frontline helpdesk assistant for a mortgage brokerage that handles customer enquiries conversationally, qualifies interest during the chat, and captures structured leads for the dedicated broker team in real time, replacing manual first-response handling.
- Architected the retrieval layer behind it with metadata-aware ingestion, hybrid retrieval and a "No Source, No Result" grounding policy, so every answer given to a customer traces back to verified source material rather than model invention. Essential in a regulated lending context.

Programming Hero | *ML Engineer* Dec 2023 – Sep 2025

- Built the Zenyora AI wellness platform (Microsoft Store), which monitors posture and continuous screen exposure through the webcam and notifies users before strain sets in, with real-time detection at 95% accuracy and sub-100ms inference.
- Added a computer-vision exercise coach that counts repetitions and corrects form on screen, plus productivity tracking that detects prolonged time in distracting applications and prompts the user to step away.
- Architected a voice-enabled RAG sales assistant over 2,000+ listings at 95% query relevance using Whisper STT and Eleven-Labs TTS.

Primacy Infotech Ltd | *AI Engineer* Jul 2023 – Nov 2023

- Led an AI tour planning platform for six UNESCO heritage sites, cutting itinerary generation from two hours to five minutes.
- Built a clustering pipeline over 5,000+ visitor surveys and presented the technical work to government stakeholders.

Rising Research Lab | *Research Assistant* Aug 2022 – May 2023

- Developed multi-disease prediction systems at 92% average accuracy, contributing to 4+ published papers.
- Implemented AutoML workflows with Optuna, reducing experimentation time by 60%.

AI FOR LOW-RESOURCE SERVICE DELIVERY

OgroPath (ogropath.com) | Founder & Lead AI Engineer

Apr 2025 – Present

- Built and scaled an AI medical entrance exam platform reaching 10,000+ students nationwide, giving learners in districts without coaching centres the same standard of preparation remotely and at low cost.
- Engineered an NLP question-generation pipeline over a 21,000+ item bank with adaptive study planning, conversational tutoring and visual learning material delivered through model APIs.
- Developed performance analysis and national ranking, identifying weak areas and generating topic-focused recommendations.

SELECTED PROJECTS

- **M-TRUST: Open-Source Fairness Toolkit** (PyPI, GitHub). Plug-in Python toolkit detecting and mitigating demographic, quality, annotation and amplification bias via a one-line API. Reduced demographic bias 30.8% with no loss of performance.
- **ExplainRAG-FC-AS: Evidence-Grounded Decision Support**. Retrieval and generation with NLI fact-checking before generation and source-level attribution. 0.85 mean evidence confidence, 90% expert preference over ranked retrieval.
- **Bengali Language Retrieval Pipeline**. LaBSE embeddings with BM25 via reciprocal rank fusion and confidence-based escalation. 95% Hit@5, 32ms response, 60% lower inference cost.
- **RAG-Based Website Generator**. Retrieval over reusable components using PostgreSQL/pgvector and AST parsing. 92% fewer tokens and 94% lower generation cost.

EDUCATION

M.Sc. in Computer Science and Engineering | BRAC University

Sep 2026 – Dec 2028 (expected)

B.Sc. in Computer Science and Engineering | University of Information Technology and Sciences (UITS)

Jan 2019 – Jun 2023

CGPA 3.62/4.00, ranked 6th in department

PUBLICATIONS & RESEARCH

12 publications · 174 citations · h-index 6 · i10-index 5 · [Google Scholar](#)

- **Nayan, N. M.**, et al. “An interpretable and balanced machine learning framework for Parkinson’s disease prediction using feature engineering and explainable AI.” *PLOS ONE*, 2025.
- Hossain, M. M., Kashem, M. A., **Nayan, N. M.**, et al. “A Medical Cyber-Physical System for Predicting Maternal Health in Developing Countries Using Machine Learning.” *Healthcare Analytics*, 2024.
- **Nayan, N. M.**, et al. “SMOTE Oversampling and Near Miss Undersampling Based Diabetes Diagnosis from Imbalanced Dataset with XAI.” *IEEE ISCC*, 2023.
- **Peer Reviewer**, *International Journal of Human–Computer Interaction* (Taylor & Francis, Q1) and *Informatics and Health* (Elsevier), 2025–Present.

Current research interests: trustworthy AI, trust and interpretability in large language models, fairness as a deployment constraint, responsible AI governance for low-resource settings.

CAPABILITY BUILDING & AI COMMUNITY

Training & Mentoring

- **AI Trainer**, Enhancing Digital Government Economy (EDGE) Project, 2023. Designed and delivered a hands-on ML/AI curriculum to 50+ participants, with 85% subsequent job placement.
- **Research Mentor**, UITS Summer Research Program, 2023 & 2024. Supervised seven students through the full research pipeline, with one Scopus-indexed paper accepted.

Events & Presentations

- **Oral Presentations**
 - IEEE Symposium on Computers and Communications (ISCC), 2023.
 - SEEDA-CECNSM, Athens, Greece, 2024.
- **Poster Presentation Coordinator**, UITS Zero One Fest, 2022.

AWARDS & RECOGNITION

- **Employee of the Month** | Primacy Infotech Ltd, 2023
- **Performance Recognition** | Robi Axiata
- **2nd Place**, Inter-university Programming Contest (45 teams) | UITS, 2022
- **Fastest Problem Solver**, Victory Day Programming Contest | UITS, 2021
- **1st Place**, Inter-university Presentation Competition | UITS, 2020

STANDARDISED TESTS

- **GRE 307/340** (Quantitative 153 · Verbal 154 · AWA 3.0)
- **TOEFL 98/120**